

```
# Credit Score Classification using Logistic Regression
```

```
# Import libraries
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import LabelEncoder
```

```
from sklearn.linear_model import LogisticRegression
```

```
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
# Create sample Credit Score dataset
```

```
data = {
```

```
    'Income': [25000, 30000, 45000, 50000, 60000, 70000, 80000, 90000,  
              35000, 40000, 55000, 65000, 75000, 85000, 28000],
```

```
    'Debt': [20000, 18000, 15000, 12000, 10000, 8000, 5000, 4000,  
            17000, 14000, 11000, 9000, 7000, 3000, 19000],
```

```
    'Payment_History': [1, 1, 0, 1, 1, 1, 1, 1,  
                       0, 1, 1, 1, 1, 1, 0],
```

```
    'Credit_Score': ['Poor', 'Poor', 'Poor', 'Good', 'Good', 'Good',  
                    'Excellent', 'Excellent', 'Poor', 'Good',  
                    'Good', 'Good', 'Excellent', 'Excellent', 'Poor']
```

```
}
```

```
df = pd.DataFrame(data)
```

```
# Display dataset
```

```
print("Credit Score Dataset:")
```

```
print(df)
```

```
# Encode target variable

encoder = LabelEncoder()

df['Credit_Score'] = encoder.fit_transform(df['Credit_Score'])


# Separate input and output

X = df[['Income', 'Debt', 'Payment_History']]
y = df['Credit_Score']


# Split dataset

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)


# Create Logistic Regression model

model = LogisticRegression(max_iter=1000)


# Train model

model.fit(X_train, y_train)


# Predict test data

y_pred = model.predict(X_test)


# Calculate accuracy

accuracy = accuracy_score(y_test, y_pred)


print("\nAccuracy:", accuracy)
print("Accuracy (%):", accuracy * 100)


# Confusion Matrix

cm = confusion_matrix(y_test, y_pred)
```

```
print("\nConfusion Matrix:")
```

```
print(cm)
```

```
# Display confusion matrix
```

```
plt.figure(figsize=(6, 4))
```

```
sns.heatmap(
```

```
    cm,
```

```
    annot=True,
```

```
    fmt='d',
```

```
    cmap='Blues'
```

```
)
```

```
plt.xlabel("Predicted")
```

```
plt.ylabel("Actual")
```

```
plt.title("Credit Score Classification - Confusion Matrix")
```

```
plt.show()
```

```
# Classification Report
```

```
print("\nClassification Report:")
```

```
print(
```

```
    classification_report(
```

```
        y_test,
```

```
        y_pred,
```

```
        target_names=encoder.classes_,
```

```
        zero_division=0
```

```
    )
```

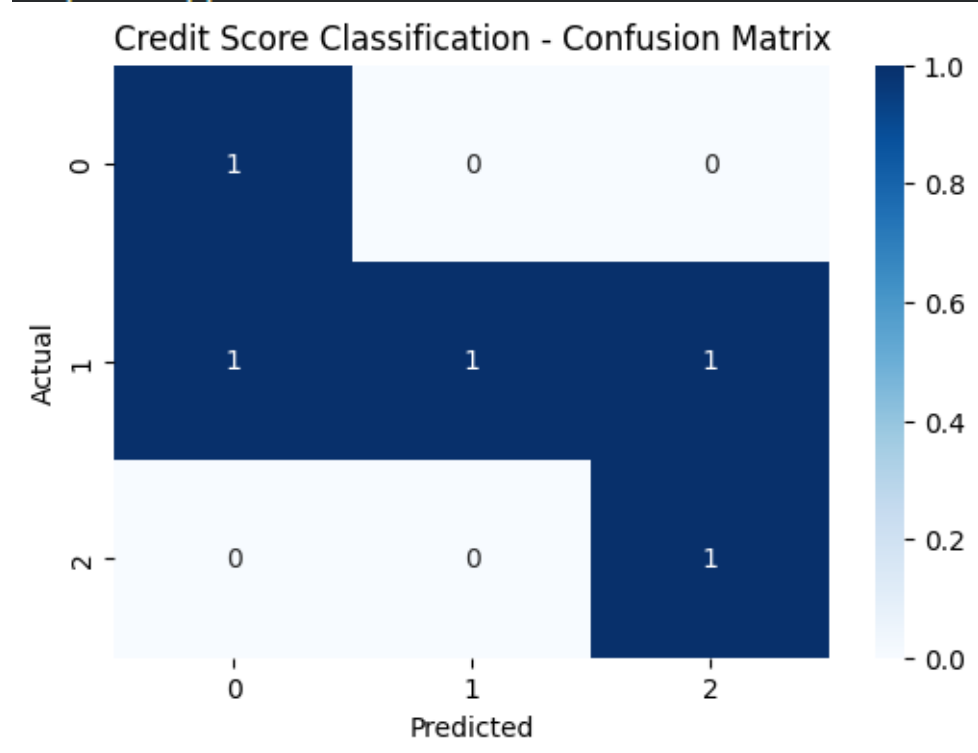
```
)
```

	Income	Debt	Payment_History	Credit_Score
0	25000	20000	1	Poor
1	30000	18000	1	Poor
2	45000	15000	0	Poor
3	50000	12000	1	Good
4	60000	10000	1	Good
5	70000	8000	1	Good
6	80000	5000	1	Excellent
7	90000	4000	1	Excellent
8	35000	17000	0	Poor
9	40000	14000	1	Good
10	55000	11000	1	Good
11	65000	9000	1	Good
12	75000	7000	1	Excellent
13	85000	3000	1	Excellent
14	28000	19000	0	Poor

Accuracy: 0.6
Accuracy (%): 60.0

Confusion Matrix:

```
[[1 0 0]
 [1 1 1]
 [0 0 1]]
```



Classification Report:

	precision	recall	f1-score	support
Excellent	0.50	1.00	0.67	1
Good	1.00	0.33	0.50	3
Poor	0.50	1.00	0.67	1
accuracy			0.60	5
macro avg	0.67	0.78	0.61	5
weighted avg	0.80	0.60	0.57	5